MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **WEEKLY CHAPTER TEST – 02 (POLYNOMIALS)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 2: Polynomials** | **MM: 20**

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, zero skipped steps, and final boxed answers with proper units.
2. In alternative (**OR**) sub-questions, complete step-by-step solutions and equal step marking apply to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Roots rejected by physical domain/restrictions are explicitly stated.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

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Q1. Which of the following curves cannot be the graph of a quadratic polynomial? [1]

(A) A parabola opening upward intersecting x-axis at 2 points.

(B) A parabola touching x-axis at exactly 1 point.

(C) A smooth continuous curve intersecting x-axis at 3 distinct points.

(D) A parabola opening downward without intersecting x-axis.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C)

We know that a quadratic polynomial $p(x) = ax^2 + bx + c$ $(a \neq 0)$ has degree 2 and:

Formula Used: Number of real zeroes of a degree-n polynomial $\leq n$

For degree 2, the graph is a parabola intersecting the x-axis in at most 2 points (0, 1, or 2). Intersecting the x-axis at 3 distinct points requires degree ≥ 3. Thus $\boxed{\text{(C)}}$ cannot be the graph of a quadratic polynomial. [1M]

Q2. If one zero of the quadratic polynomial $x^2 + 3x + k$ is 2, then the value of k is: [1]

(A) 10 (B) -10 (C) -7 (D) -2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B)

Let $p(x) = x^2 + 3x + k$. We know that if $x = \alpha$ is a zero of $p(x)$, then:

Formula Used: $p(\alpha) = 0$

Substituting $x = 2$ into $p(x) = 0$:

$$p(2) = (2)^2 + 3(2) + k = 0 \implies 4 + 6 + k = 0 \implies 10 + k = 0 \implies \boxed{k = -10}. \quad [1M]$$

Q3. A quadratic polynomial, the sum of whose zeroes is 0 and one zero is 3, is: [1]

(A) $x^2 - 9$ (B) $x^2 + 9$ (C) $x^2 + 3$ (D) $x^2 - 3$

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A)

Let the two zeroes be $\alpha = 3$ and β. Given $\alpha + \beta = 0 \implies 3 + \beta = 0 \implies \beta = -3$. We know the standard quadratic polynomial form:

Formula Used: $p(x) = k\big(x^2 - (\alpha + \beta)x + \alpha\beta\big)$

Here $\alpha + \beta = 0$ and $\alpha\beta = (3)(-3) = -9$. [0.5M]

Taking $k = 1$ yields $p(x) = x^2 - 0 \cdot x - 9 = \boxed{x^2 - 9}$. [0.5M]

Q4. A quadratic polynomial, the sum of whose zeroes is -5 and their product is 6, is: [1]

(A) $x^2 + 5x + 6$ (B) $x^2 - 5x + 6$ (C) $x^2 - 5x - 6$ (D) $-x^2 + 5x + 6$

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A)

Given $S = \alpha + \beta = -5$ and $P = \alpha\beta = 6$. We know the quadratic formula in terms of S and P:

Formula Used: $p(x) = k\big(x^2 - Sx + P\big)$

Substituting $S = -5$ and $P = 6$ with $k = 1$:

$$p(x) = x^2 - (-5)x + 6 = \boxed{x^2 + 5x + 6}. \qquad [1M]$$

Q5. Assertion-Reason: [1]

Assertion (A): The zeroes of the quadratic polynomial $x^2 + 99x + 127$ are both negative.

Reason (R): If all coefficients a, b, c in $ax^2 + bx + c$ have the same positive sign, then all real zeroes must be negative.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A) Both A and R are true and R is the correct explanation of A.

We know the discriminant and Vieta's sign relations for $ax^2 + bx + c$ $(a, b, c > 0)$:

Formula Used: $D = b^2 - 4ac, \qquad \alpha + \beta = -\dfrac{b}{a} < 0, \qquad \alpha\beta = \dfrac{c}{a} > 0$

For $x^2 + 99x + 127$: $D = 99^2 - 4(1)(127) = 9801 - 508 = 9293 > 0$, confirming two distinct real zeroes. Since $\alpha\beta = 127 > 0$ (both same sign) and $\alpha + \beta = -99 < 0$, both zeroes must be negative. Moreover, for any $a, b, c > 0$ with $D \geq 0$, every real zero satisfies the same sign logic. Both **(A)** and **(R)** are true and **(R)** explains **(A)**. [1M]

Q6. Find the zeroes of the quadratic polynomial $p(t) = t^2 - 15$ and verify the relationship between zeroes and coefficients. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Comparing $p(t) = t^2 + 0 \cdot t - 15$ with $at^2 + bt + c$, we have $a = 1$, $b = 0$, $c = -15$. We know the algebraic difference-of-squares identity and Vieta's relations:

Formula Used: $x^2 - y^2 = (x - y)(x + y), \qquad \alpha + \beta = -\frac{b}{a}, \qquad \alpha\beta = \frac{c}{a}$

Step 1 – Finding zeroes: Writing $15 = (\sqrt{15})^2$:

$$p(t) = t^2 - (\sqrt{15})^2 = (t - \sqrt{15})(t + \sqrt{15}) = 0 \implies \alpha = \boxed{\sqrt{15}},\ \beta = \boxed{-\sqrt{15}}. \qquad \textbf{[1M]}$$

Step 2 – Verification:

$$\alpha + \beta = \sqrt{15} + (-\sqrt{15}) = 0, \qquad -\frac{b}{a} = -\frac{0}{1} = 0 \implies \alpha + \beta = -\frac{b}{a}, \qquad \textbf{[0.5M]}$$

$$\alpha\beta = (\sqrt{15})(-\sqrt{15}) = -15, \qquad \frac{c}{a} = \frac{-15}{1} = -15 \implies \alpha\beta = \frac{c}{a}. \qquad \textbf{[0.5M]}$$

Q7. If α and β are the zeroes of the polynomial $p(y) = 5y^2 - 7y + 1$, find the value of: $\frac{1}{\alpha} + \frac{1}{\beta}$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Comparing $p(y) = 5y^2 - 7y + 1$ with $ay^2 + by + c$, we have $a = 5$, $b = -7$, and $c = 1$. We know Vieta's sum and product of zeroes:

Formula Used: $\alpha + \beta = -\frac{b}{a}, \qquad \alpha\beta = \frac{c}{a}$

Substituting $a = 5$, $b = -7$, $c = 1$:

$$\alpha + \beta = -\frac{(-7)}{5} = \frac{7}{5}, \qquad \alpha\beta = \frac{1}{5}. \qquad \textbf{[1M]}$$

Combining $\frac{1}{\alpha} + \frac{1}{\beta}$ over their common denominator $\alpha\beta$:

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\beta + \alpha}{\alpha\beta} \qquad \textbf{[1M]} = \frac{7/5}{1/5} = \frac{7}{5} \times \frac{5}{1} = \boxed{7}. \qquad \textbf{[1M]}$$

Q8. If $x = 1$ is a zero of the polynomial $p(x) = ax^2 - 3(a - 1)x - 1$, then determine the value of a. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given that $x = 1$ is a zero of $p(x) = ax^2 - 3(a - 1)x - 1$. We know from the definition of a zero of a polynomial:

Formula Used: $x = k$ is a zero of $p(x) \implies p(k) = 0$

Substituting $x = 1$ into $p(x)$ and equating to zero:

$$p(1) = 0 \implies a(1)^2 - 3(a - 1)(1) - 1 = 0 \qquad \textbf{[1M]}$$

$$\implies a - 3a + 3 - 1 = 0 \qquad \textbf{[1M]}$$

$$\implies -2a + 2 = 0 \implies 2a = 2 \implies \boxed{a = 1}. \qquad \textbf{[1M]}$$

Q9. For what value of k is -4 a zero of the quadratic polynomial $x^2 - x - (2k+2)$? **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let $p(x) = x^2 - x - (2k+2)$. Since $x = -4$ is a zero of $p(x)$, we know:

Formula Used: $p(-4) = 0$

Substituting $x = -4$ step-by-step:

$$p(-4) = 0 \implies (-4)^2 - (-4) - (2k+2) = 0 \quad \text{[1M]}$$
$$\implies 16 + 4 - 2k - 2 = 0 \quad \text{[1M]}$$
$$\implies 18 - 2k = 0 \implies 2k = 18 \implies \boxed{k = 9}. \quad \text{[1M]}$$

SECTION C: Applied Case Study Analysis

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Q10. Case Study: Storm Damage Bent Wire

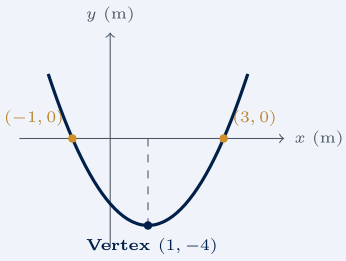
A metal electrical cable got bent into a parabolic profile intersecting the ground (x-axis) at $x = -1$ and $x = 3$, with its lowest point at vertex $(1, -4)$ (scale 1 unit = 1 m). **[4]**

(i) Name the mathematical curve formed by the bent wire and state the number of zeroes of the representing polynomial. **[1]**

(ii) Write the zeroes of the polynomial represented by the shape of the wire. **[1]**

(iii) **(a)** Determine the quadratic polynomial expression $f(x)$ representing the shape of the wire. **OR** **(b)** Find the value of the polynomial at $x = 4$ and at $x = -2$. What do these values indicate about the axis of symmetry? **[2]**

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]



Graphical Profile: Cuts ground line $y = 0$ at $(-1, 0)$ and $(3, 0)$, with lowest point at vertex $(1, -4)$ determining $a = 1$ uniquely.

Formula Used: $f(x) = a(x-\alpha)(x-\beta) = a\big(x^2 - (\alpha+\beta)x + \alpha\beta\big)$

(i) Curve & zeroes count: An $\boxed{\text{upward-opening parabola}}$ crossing the x-axis twice, so it has $\boxed{2 \text{ zeroes}}$. [1M]

(ii) Zeroes: From the x-axis intercepts $(-1, 0)$ and $(3, 0)$, the zeroes are $x = \boxed{-1}$ and $x = \boxed{3}$. [1M]

(iii)(a) Quadratic polynomial $f(x)$: With $\alpha = -1, \beta = 3$, $f(x) = a(x+1)(x-3) = a(x^2 - 2x - 3)$. Using vertex $(1, -4)$:

$$f(1) = -4 \implies a(1+1)(1-3) = -4 \implies -4a = -4 \implies a = 1 \implies \boxed{f(x) = x^2 - 2x - 3}. \quad \text{[2M]}$$

OR (Alternative Part (b) – Symmetry check): At $x = 4$ and $x = -2$ in $f(x) = x^2 - 2x - 3$:

$$f(4) = 4^2 - 2(4) - 3 = \boxed{5 \text{ m}}, \qquad f(-2) = (-2)^2 - 2(-2) - 3 = \boxed{5 \text{ m}}. \quad \text{[1M]}$$

Equal values at $x = 4$ and $x = -2$ (both 3 units from $x = 1$) show $\boxed{x = 1 \text{ is the axis of symmetry.}}$ [1M]

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★